

REBANTA ROY

Redmond, WA | rr3659@nyu.edu | +1(347)-200-2757 | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Engineer with 5 years of experience across Azure AI/ML, enterprise automation, and cloud engineering. Specialized in Agentic systems, correlation systems, observability platforms and analyzing data, and production-grade automation on Azure.

EDUCATION

M.S. Computer Science and Data Science | Westcliff University, CA | GPA: 4.0/4.0 **May 2024 – Present**

Coursework: Data Engineering, Data Science, Dynamic Programming, Software Development, Cloud Architecture

M.S. Robotics and Machine Learning | New York University, NY | GPA: 3.50/4.0 **2019 – 2021**

Coursework: Machine Learning, Robot Perception, Computer Vision, Deep Learning, App Development, DevOps, and Deployment

B.Tech. Mechanical Engineering and Automation | KIIT University, India | GPA: 3.64/4.0 **2015 – 2019**

Coursework: Programming in C, OOP in C++, Numerical Methods, Mechatronics, Robotics, Automation

TECHNICAL SKILLS

Agentic AI / Automation: GitHub Copilot Agents, SRE Agent, Copilot Studio Agent, Power Automate, MCP, Multi-agent orchestration, Asynchronous workflows, REST API orchestration, OAuth2, RBAC, audit logging, rate limiting.

ML / AI: Transformers, Bi-LSTM, GANs, CNN/RCNN, ResNet, attention mechanisms, vectorization/embeddings, clustering, HDBSCAN, LightGBM, XGBoost-gpu, fuzzy string matching, NLP pattern systems, curve fitting/optimization, power-law models.

Azure: Data Factory, Synapse Analytics, Azure Functions, Azure Data Explorer, ADLS Gen2, Storage, VM, Networking, App Service, Application Insights, Azure Monitor, Log Analytics, Key Vault, SQL Database, Managed Identity, Entra ID, Dataverse, Kubernetes.

Data / Analytics: Python, Pandas, NumPy, SciPy, scikit-learn, TensorFlow, Keras, Matplotlib, Sentence-transformers, SQL, KQL.

Engineering Tools: Git, Linux, Docker, PowerShell, Bash, Azure CLI, Jupyter, OpenAPI/Swagger, Gunicorn, VS Code,

Azure DevOps: Deployments, Storage Synchronization, Repos, Pipelines, Work Items, Wikis, Test Plans.

Programming: Python, C++, SQL, Node.js, TypeScript, React, Flask, HTML/CSS/JavaScript.

Robotics / CV: OpenCV, SLAM, semantic segmentation, trajectory planning & generation, frame tracking, Grad-CAM.

PROFESSIONAL EXPERIENCE

AI and MLOps Architect | LTIMindtree Ltd. | Bellevue, WA, USA

December 2025 – Present

Project: Compliance Orchestrator Agent & Rack-Tools MCP Plugins

Built a GitHub Copilot agent on the Agency CLI that orchestrates parallel ADO pipelines from tickets, evaluates artifacts, and publishes results, packaged as a pip-installable plugin with a custom MCP server, agent files, and an automated eval pipeline.

- Designed an end-to-end orchestration flow integrating Ticketing MCP, custom rack-status MCP, and ADO MCP to fetch incidents, resolve rack-to-host mappings, trigger pipelines, poll execution status, and download artifacts in parallel.
- Developed a custom rack-status MCP server in TypeScript to expose rack inventory lookup, batch host resolution, and per-host unmatched parsing across staging and report folders.
- Authored six declarative agent.md workflows, including compliance reviewer, pipeline triage, and PR reviewer agents, with scoped MCP tool permissions and YAML-validated frontmatter.
- Packaged the solution as a pip-installable wheel and Azure Artifacts Universal Package, enabling reusable deployment across teams and environments.
- Built an automated Azure DevOps evaluation pipeline to regression-test agent behavior, plugin changes, workflow routing, and compliance output quality on every update.

Project: End-to-End MLOps Pipeline for Inference Benchmarking

Designed a production Azure DevOps pipeline that runs Mistral inference benchmarks, covering host pre-check, VM lifecycle, driver and SDK provisioning, parallel execution across accelerators, and structured KPI publication.

- Built production Azure DevOps automation for accelerator validation workflows, covering host readiness checks, VM lifecycle management, remote execution, software provisioning, and structured benchmark artifact generation.
- Engineered reliable driver and device lifecycle handling to ensure repeatable accelerator availability across runs, with automated validation, KPI extraction, and dashboard-ready CSV/JSON outputs.
- Improved pipeline resilience by adding guarded diagnostics, failure-safe cleanup, reusable PowerShell modules, backward-compatible updates, and standardized teardown logic to prevent dirty host states.

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- Standardized shared automation modules and execution templates to support reuse across multiple validation workflows while reducing manual setup and maintenance effort.

AI Developer | LTIMindtree Ltd. | Bellevue, WA, USA

Mar 2023 – December 2025

Project: Agentic TSG Error Classification System

Built a 2-agent closed-loop platform that classifies production ADF/Synapse errors, maps each to a fix workflow, and continuously refines rules via validation feedback. Scaled to 164 categories using 350+ rules and normalization patterns.

- Designed Agent 1 to build message skeletons, unwrap nested error structures, and rank candidate rules by key-phrase extraction into regex mapping JSON.
- Agent 2 runs validation passes, generates TSG-linked fix suggestions, and routes mismatches back for closed-loop refinement.
- Engineered a 350+ rule classification engine across 164 categories with regex matching and 50+ normalization patterns, achieving full coverage on production ADF/Synapse error data.
- Added feedback-driven refinement workflows to track validation mismatches, improve rule accuracy, and continuously strengthen automated error-to-resolution mapping.

Project: ADF Pipeline Observability Platform and Ticket Generation

Designed and deployed an ADF observability platform for deep validation, health monitoring, and adaptive long-running detection across 30+ production Data Factories. Delivered as a serverless Azure Function App with resumable ingestion and reporting.

- Built monitoring outputs with interactive HTML dashboards and multi-sheet Excel reports with direct ADF Studio deep links, based on overall health check and deep validation of pipelines.
- Developed a power-law threshold model using SciPy curve fitting (L-BFGS-B) to replace static long-running thresholds with adaptive detection. Deployed as Azure Functions using HTTP triggers, async request-reply, checkpoint-resumable pagination, and Managed Identity.
- Based on power-law threshold, Long Running pipelines were determined, and Power Automate Flow was created based on failed categories tracked in Dataverse. Utilized API to create, update, and mitigate tickets in queue based on inputs.
- Automated end-to-end incident creation and mitigation workflows by connecting pipeline health signals, failure categorization, Dataverse tracking, and ticket lifecycle updates.

Project: Enterprise MCP Hub & GitHub Copilot Agent Platform

Architected a unified MCP aggregator connecting 10+ enterprise backends into a single tool gateway for GitHub Copilot, built 11 purpose-built agents on top of it, and implemented enterprise security, adoption telemetry, and agent lifecycle governance.

- Unified 100+ tools across Azure DevOps, Ticketing, Kusto, Metrics Platform, SharePoint, ARM/Bicep, Azure Monitor, Email, Filesystem, and custom Python analyzers through a TypeScript aggregator with dynamic server discovery and protocol-level routing with reusable governance patterns to standardize agent logs, tool access, telemetry tracking, and operations.
- Implemented an 8-module security stack with RBAC, token-bucket rate limiting, cryptographic session management, audit logging with rotation, and automated token renewal.
- Defined 11 agents using declarative agent.md workflows with tool restrictions and scoped instructions and built ADO-integrated PR auditing and checklist generation using 80+ Azure DevOps MCP tools.
- Implemented JSONL telemetry logging with deduped report sync based on unique IDs and Power BI dashboards and enforced a 4-step agent lifecycle with compliance validation and onboarding wizard support.

Project: Copilot Studio Agent & Synapse Analytics Middleware

Built a Copilot Studio Agent with Child Agents to orchestrate ADF Pipeline runs based on Ticketing Center, ADE and SQL inputs. Flask REST API middleware layer was created on Azure App Service for Synapse operations, published as OpenAPI for Copilot Studio integration via OAuth2.

- Copilot Studio Agent was created with 5 Child Agents in order to retrieve details from queues, figure out pipeline names, and find out details from contained Subscription.
- Kusto Connector and SQL connector were utilized in Child Agents, to gather details related to current status, and necessary actions were taken from ADF Connectors and integrated Synapse Web App.
- Built a Flask REST API (27 endpoints) enabling SQL script CRUD, parameterized execution, Spark pooling, and notebook management with layered Azure AD security.
- Published an OpenAPI 3.0.3 spec enabling Copilot Studio integration via OAuth 2.0 authorization code flow.

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- Enabled secure natural-language-driven operations by connecting Copilot workflows with backend APIs, enterprise authentication, and governed execution paths for Synapse and pipeline actions.

Project: SRE Agent to Monitor Internal Incident Management System and Control Pipeline Runs

Developed an automation suite on top of Microsoft SRE Agent to correlate incidents to ADF pipelines using Kusto + fuzzy matching, expose operations through REST and Connectors, and automate pipeline reruns.

- Built a Ticketing Service to ADF correlation script on SRE Agent using Kusto queries, JSON parsing, and 4-layer fuzzy matching with confidence scoring.
- Ticket details were parsed and compared against Dataverse tables for internal Analysis
- Built the Instruction Script for SRE agent integrating Kusto, Log Analytics, and ADF APIs to interpret errors and autonomously restart failed pipelines.
- Added validation and fallback logic to reduce false matches, improve incident-to-pipeline mapping accuracy, and ensure only high-confidence failures were selected for automated recovery actions.

Project: Data Lineage Visualization Engine

Built a Hierarchy determining System to track ADW and ADLS objects. Over 7000 objects hierarchies were determined with validated pipeline-to-hierarchy mapping.

- Implemented interactive cross-linked filters across source, transformation, and serving layers with pipeline-to-hierarchy validation sourced from Kusto and SQL metadata.

Project: DRI Handbook Migration

Migrated a 344-page OneNote DRI handbook to Git-based Markdown with automated conversion, extraction, and sanitization.

- Built a Python conversion pipeline handling DOCX parsing, embedded image extraction, code block language auto-detection, and pre-commit hooks for credential sanitization across 344 pages.

Azure Infrastructure Team Lead | LTIMindtree Ltd. | Bellevue, WA, USA

Sep 2022 – Mar 2023

Project: Azure Hybrid Infrastructure & Network Operations

Led Azure network engineering for Fortune 500 clients: break-fix, hybrid connectivity, VMware-to-Azure migrations, and monitoring across VMs, VNETs, ExpressRoute, VPN, and Firewall.

- Managed Premier/Pro Direct SLA queues for VMs, VNETs, ExpressRoute, VPN, LBs, NSGs, and Firewall; resolved high-severity hybrid connectivity issues (BGP routing, IPsec/IKE, site-to-site tunnels); SME for NVAs, Route Server, Virtual WAN, and Bastion.
- Automated provisioning with PowerShell, Azure CLI, ARM/Bicep, and Runbooks; led VMware-to-Azure migrations using Azure Migrate and hybrid vSphere integration.
- Drove proactive monitoring with Network Watcher, Log Analytics, and Azure Monitor; performed RCA, authored HLD/LLD/SOP docs, and mentored junior engineers.

Azure AI and Machine Learning Team Lead | LTIMindtree Ltd. | Bellevue, WA, USA

Feb 2022 – Sep 2022

Project: Azure AI/ML Enterprise Platform Delivery

Led 12 engineers delivering Azure Cognitive Services, ML Studio, and IoT Hub solutions for enterprise clients across API integration, distributed training, AKS deployment, and production pipeline automation.

- Implemented Vision, Speech, Language, and Decision APIs via REST/SDK (Python, C++, .NET), managed distributed training, AKS deployments, model versioning, and monitoring in Azure ML workspaces.
- Delivered RCA for AutoML, pipeline, and inferencing failures; partnered with DevOps, Security, and IoT Edge teams for optimized deployments and A/B testing.
- Automated ML pipeline deployment and post-deployment monitoring using PowerShell and Azure CLI.

Azure Cognitive Services / Machine Learning Engineer | LTIMindtree Ltd. | Bellevue, WA, USA

Aug 2021 – Mar 2022

Project: Azure AI & IoT Service Delivery

Supported enterprise customers on Azure Cognitive Services, ML, and IoT: model deployment, IoT integration, ML pipeline optimization, and production availability.

- Delivered REST/SDK integrations (Python, C++, .NET) for Vision, Speech, Language, and Decision APIs; deployed ML models in Azure ML Studio with autoscaling, failover, and compute optimization.

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- Supported IoT Hub, IoT Edge, device twins, and stream analytics for real-time data and anomaly detection, automated production ML pipelines for improved efficiency and scale.

Research Assistant | Systems and Automatic Controls Lab, NYU | Brooklyn, NY, USA

Sep 2020 – May 2022

Project: Control Systems Research & Lab Automation

Researched automation with real-time C++ data acquisition via DAQ and PID/LQR controller design with stability analysis.

- Built real-time C++ acquisition pipelines with NI DAQ units for signal capture, filtering, and hardware-in-the-loop experiments, validated control methods for improved system efficacy.
- Designed PID and LQR controllers with modeling and stability analysis for automation and dynamic systems.

ACADEMIC PROJECTS

Project: Pattern Recognition of Human Speech Stuttering | NYU, Dept. of CSE

- Curated and structured UCLASS Children Speech Database, audio data was organized pertaining to Regions of Interest for distinct stuttering patterns, and the data was labelled according to their patterns.
- Preprocessed raw audio by changing sample rates, feature scaling, and windowing followed by feature extraction using Squeeze and Excitation ResNet architecture, along with inversion of frequency and melspectrography.
- Addressed class imbalance and trained models using Bi-LSTM, Transformers, KNN, and Ensemble classifiers, achieving a 92% testing accuracy with a Visual Transformer integrated with an SE-ResNet and Multi-head Attention mechanism.

Project: Classification of Network Traffic | NYU, Dept. of CSE

- Created a deep learning model that classifies network traffic in different categories such as Gaming, VoIP and Video Streaming using Wireshark packet captures.
- Engineered key traffic features included packet size, inter-arrival time, flow duration, and protocol types such as TCP window size and control flags.
- Developed and optimized a CNN model which was trained using CIC-IDS2017 dataset, live packet captures were used for testing, and the model was deployed as a RESTful API using Flask for real time traffic classification.
- The CNN model proved the best architecture, and it produced 86% accuracy.

Project: Scoring and Classifying Activities | NYU, Dept. of CSE

- Created a custom JSON database containing Sports and Dance videos found on YouTube based on desired parameters such as clear poses and the visibility of body joints.
- Created a hybrid model of VGG16, VGG19 and custom CRNN (Convolutional Recurrent Neural Network) to classify activities and achieved an accuracy of 97% along with scoring the performance.

Project: Extracting Location Data from Satellite Images | NYU, Dept. of CSE

- Created a feature matching pipeline for location identification which includes the extraction of key points from images and subsequently matching them with blocks of satellite images.
- Used feature vector similarity based fast image retrieval and precise localization through Flann based k-d tree optimization with Decision Trees.

Project: Classification of Urban Noise | NYU, Dept. of CSE

- Improved the quality of urban sound datasets by extracting features using MFCC and applying Gaussian noise filtering during preprocessing stages.
- Used fully connected (FC), CNN, and RCNN models for audio classification and validation on test set resulted in 89% accuracy, 87% testing accuracy with 10% false detection in k-fold cross-validation. The model with RCNN 16 layers provided the best results, guaranteeing the necessity of considering frames of audio.

VOLUNTEER EXPERIENCE

Volunteer Full-Stack Web Developer | Uttoron | Bellevue, USA

Oct 2024 – Present

- Built a full-stack web platform (React, Node.js) with registration, admin roles, donations, events, membership management, and voting rights.
- Deployed on Azure (Storage, SQL DB, Key Vault, Front Door, Web App, VNet) with PayPal payments, email notifications, and role-based access control.